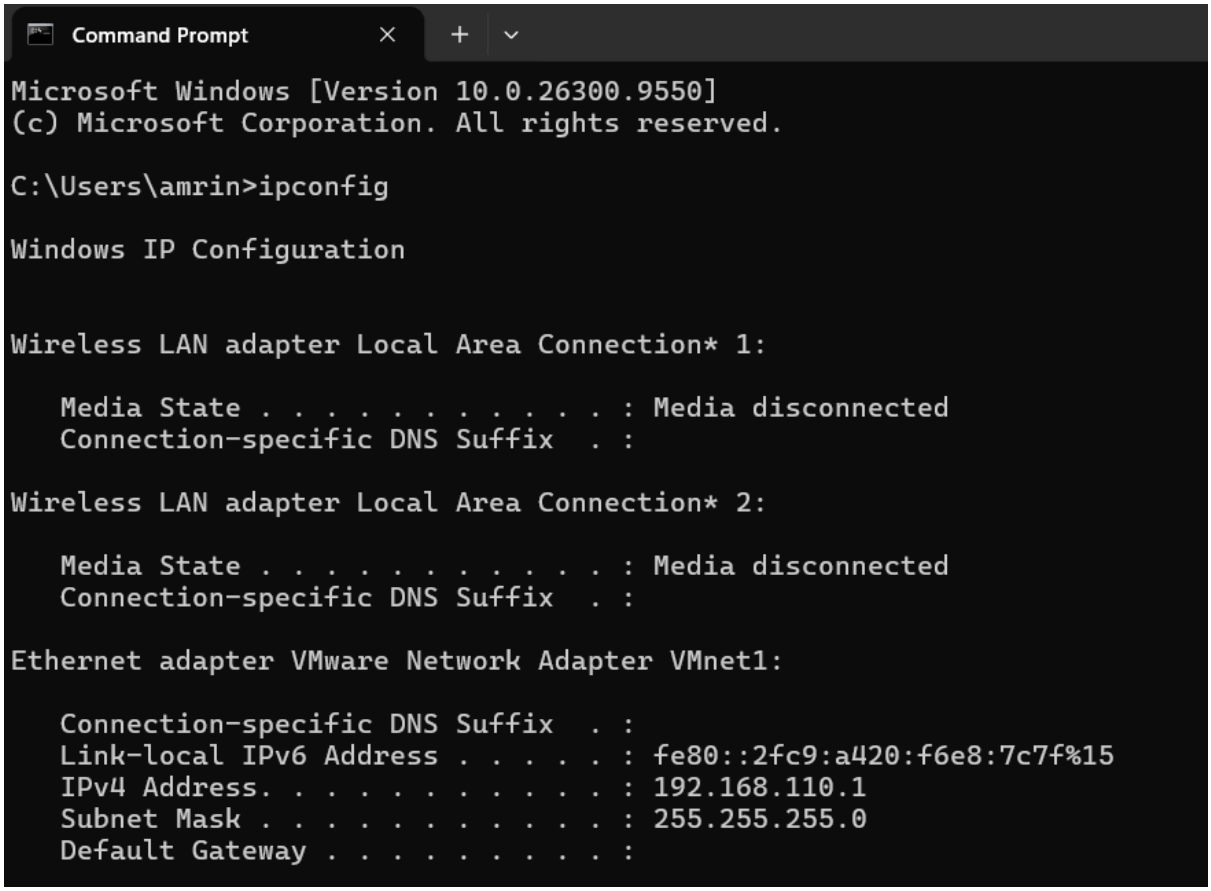
Assignment 10

1.Find the IP address of your computer or mobile device and write down whether it is IPv4 or IPv6. Also mention if it is a public or private IP.

Ans-



**2.Given the following list of IP addresses, identify which ones are valid private IPs and which are public: 192.168.1.10, 10.0.0.5, 172.16.5.20, 8.8.8.8, 123.45.67.89.

Hint: Check the standard private IP address ranges for IPv4.**

Ans-

Standard private IPv4 ranges:

Private Range	Example
10.0.0.0 – 10.255.255.255	10.0.0.5
172.16.0.0 – 172.31.255.255	172.16.5.20
192.168.0.0 – 192.168.255.255	192.168.1.10

IP Address	Type	Reason
192.168.1.10	Private	Inside 192.168.0.0/16
10.0.0.5	Private	Inside 10.0.0.0/8
172.16.5.20	Private	Inside 172.16.0.0/12
8.8.8.8	Public	Not in private ranges
123.45.67.89	Public	Not in private ranges

**3.Write a short script in Python or JavaScript that takes an IP address as input and tells the user whether it is a valid IPv4 address or not.

Constraint: Do not use any third-party libraries for validation.**

Ans-

Step 1: Install/open Python

Open IDLE, VS Code, or any Python editor.

Step 2: Create a file

Create: ipv4_validator.py

Step 3: Enter this code

```
ip = input("Enter an IP address: ")
parts = ip.split(".")
valid = True
if len(parts) != 4:
    valid = False
else:
    for part in parts:
        if not part.isdigit():
            valid = False
            break
        number = int(part)
        if number < 0 or number > 255:
            valid = False
            break
```

```
if valid:
    print("Valid IPv4 address")
else:
    print("Invalid IPv4 address")
```

Step 4: Run the program

Enter an IP address: 192.168.1.300

Invalid IPv4 address

4.Explain with an example how Instagram or WhatsApp might use IP addresses to detect suspicious login activity from different locations.

Ans-

Step 1 — Normal login

You log in:

Phone

↓

Wi-Fi/ISP

↓

Internet

↓

Instagram Server

Instagram can see the public IP associated with the connection.

Step 2 — IP is recorded

The service can associate the login with information such as:

IP address

Approximate location

Login time

Device information

Step 3 — New login occurs

Someone logs into the same account from another network:

New IP → Different approximate location

For example:

Login 1:

India

10:00 AM

Login 2:

United Kingdom

10:15 AM

Step 4 — Security system compares activity

The service can compare the new login with previous login patterns.

A sudden unusual location/IP can contribute to a risk score.

Step 5 — Security action

Depending on the service's security system, it may:

- Send a login alert
- Ask for verification
- Require a password/OTP
- Ask the user to confirm the login
- Temporarily restrict suspicious activity